PRACTICE EXAM 02 / EXAM SET 01

Domain 2: Anatomy and Physiology

8 scored questions	4 answer choices per question	Answer key + explanations included

Instructions

Choose the single best answer (A, B, C, or D) for each question. Mark your responses before checking the answer key and detailed explanations at the end of this document. These practice questions are not official NHA examination items.

QUESTIONS

Q01 | Body Structures and Organ Systems

> **SCENARIO** While reviewing a provider-approved heart diagram, a patient says, "A vein must carry blood that has very little oxygen." The medical assistant uses the pulmonary circulation to clarify this statement.

Which description correctly explains the role of the pulmonary veins?

A. They carry oxygen-poor blood from the right ventricle to the lungs.
B. They carry oxygen-rich blood from the left ventricle to the body.
C. They carry oxygen-rich blood from the lungs to the left atrium.
D. They carry oxygen-poor blood from the body to the right atrium.

Q02 | Body Structures and Organ Systems

> **SCENARIO** An older adult with chronic kidney disease has provider-documented anemia attributed to reduced erythropoietin production. The medical assistant is reviewing the record before a follow-up visit.

Which normal interaction between body systems is reduced in this patient?

A. The kidneys release a hormone that stimulates red blood cell production in bone marrow.
B. The bone marrow releases a hormone that increases urine production in the kidneys.
C. The kidneys release a hormone that stimulates platelet production in the spleen.
D. The spleen releases a hormone that increases oxygen absorption in the lungs.

Q03 | Pathophysiology and Disease Processes

> **SCENARIO** A child with established asthma returns for a routine follow-up. The parent asks why the child sometimes has trouble breathing out when asthma symptoms occur. The medical assistant reinforces the provider's explanation.

Which change most directly explains this difficulty?

A. The pleural cavity fills with air and compresses the lung.

B. Fluid collects in the alveoli because the left ventricle fails.

C. A clot blocks blood flow through a pulmonary artery.

D. Airway inflammation and muscle tightening narrow the breathing passages.

Q04 | Body Structures and Organ Systems

> **SCENARIO** A patient had a large portion of the small intestine surgically removed. The provider has arranged nutritional follow-up because of concern about reduced absorption of nutrients from food.

Which normal function of the removed organ best explains this follow-up?

A. It stores bile until fat enters the digestive tract.

B. It absorbs most nutrients after food has been digested.

C. It produces the acid that begins protein digestion.

D. It moves swallowed food from the throat to the stomach.

Q05 | Pathophysiology and Disease Processes

> **SCENARIO** An adolescent with newly diagnosed type 1 diabetes asks why prescribed insulin is needed. The provider has explained that the immune system damaged the pancreatic cells that produce insulin. The medical assistant reinforces this teaching.

Which statement best connects the disease process with the purpose of insulin?

A. Insulin replaces digestive enzymes that normally break down carbohydrates in the intestine.

B. Insulin replaces a missing hormone that helps glucose enter many body cells.

C. Insulin prevents the kidneys from making glucose during urine formation.

D. Insulin replaces a hormone that normally releases stored glucose from the liver.

Q06 | Body Structures and Organ Systems

SCENARIO A patient scheduled for cystoscopy asks how the provider can reach the bladder without making an abdominal incision. The medical assistant reviews the provider-approved procedure handout, which describes passage through the normal opening used to urinate.

Which structure provides this route from outside the body to the bladder?

A. Ureter
B. Renal pelvis
C. Urethra
D. Renal artery

Q07 | Pathophysiology and Disease Processes

SCENARIO During intake, a patient reports several days of very limited walking while wearing a lower-leg cast and new one-sided calf swelling. The medical assistant promptly reports both findings to the provider, who is evaluating for a possible deep vein thrombosis.

Which explanation correctly distinguishes a risk factor from a possible manifestation?

A. Limited walking is a manifestation because it proves that a vein is blocked.
B. Calf swelling is a risk factor because it slows blood before a clot forms.
C. The cast rules out a clot because compression always increases venous flow.
D. Limited walking is a risk factor because reduced muscle activity can slow venous blood flow.

Q08 | Pathophysiology and Disease Processes

Which physiological change best explains why a person with hypothyroidism may feel unusually cold?

A. Excess thyroid hormone increases heat production.
B. Excess thyroid hormone accelerates intestinal movement.
C. Insufficient thyroid hormone reduces metabolic heat production.
D. Insufficient thyroid hormone increases metabolic energy use.

ANSWER KEY & EXPLANATIONS

Practice Exam 02 | Domain 2: Anatomy and Physiology | 8 scored questions

01 C	02 A	03 D	04 B	05 B	06 C
07 D	08 C				

DETAILED EXPLANATIONS

Q01 | Correct answer: C

They carry oxygen-rich blood from the lungs to the left atrium.

Pulmonary veins return oxygen-rich blood from the lungs to the left atrium. A vessel is classified as a vein because it carries blood toward the heart, rather than because of its oxygen content. The pulmonary arteries carry oxygen-poor blood from the right ventricle to the lungs.

Q02 | Correct answer: A

The kidneys release a hormone that stimulates red blood cell production in bone marrow.

The kidneys normally produce erythropoietin, which signals bone marrow to produce red blood cells. Reduced erythropoietin can therefore contribute to anemia in chronic kidney disease. This links renal endocrine function with blood cell production; it does not mean that bone marrow controls urine production.

Q03 | Correct answer: D

Airway inflammation and muscle tightening narrow the breathing passages.

Asthma can cause airway swelling, increased mucus, and tightening of the muscles around the airways. These changes narrow the passages and limit airflow, including airflow during exhalation. Air in the pleural cavity describes a different mechanism and is not the usual explanation for asthma symptoms.

Q04 | Correct answer: B

It absorbs most nutrients after food has been digested.

The small intestine is the main site where nutrients from digested food are absorbed. Removing a large portion can reduce the available absorptive area, which explains the need for nutritional follow-up. Bile storage occurs in the gallbladder, while acid production occurs in the stomach.

Q05 | Correct answer: B

Insulin replaces a missing hormone that helps glucose enter many body cells.

Autoimmune destruction of pancreatic insulin-producing cells causes the insulin deficiency of type 1 diabetes. Prescribed insulin replaces this hormone and helps regulate blood glucose, including by promoting glucose entry into muscle and fat cells. Digestive enzymes have a different function, and a hormone that promotes glucose release describes glucagon rather than insulin.

Q06 | Correct answer: C

Urethra

The urethra connects the bladder to the outside of the body and provides the usual access route for cystoscopy. The ureters instead carry urine from the kidneys to the bladder. Recognizing this route explains the anatomy of the ordered procedure without requiring the assistant to perform it.

Q07 | Correct answer: D

Limited walking is a risk factor because reduced muscle activity can slow venous blood flow.

Reduced movement decreases the assistance that leg muscle contractions provide to venous return, promoting blood stasis and increasing clot risk. New one-sided swelling is a possible manifestation that requires evaluation, rather than proof of a diagnosis. A cast does not rule out thrombosis, and the assistant should not diagnose the cause of the swelling.

Q08 | Correct answer: C

Insufficient thyroid hormone reduces metabolic heat production.

Thyroid hormones help regulate the rate at which the body uses energy. When too little thyroid hormone is available, metabolic activity and heat production can decrease, contributing to cold intolerance. Increased metabolic energy use would fit the opposite direction of change.

End of Practice Exam